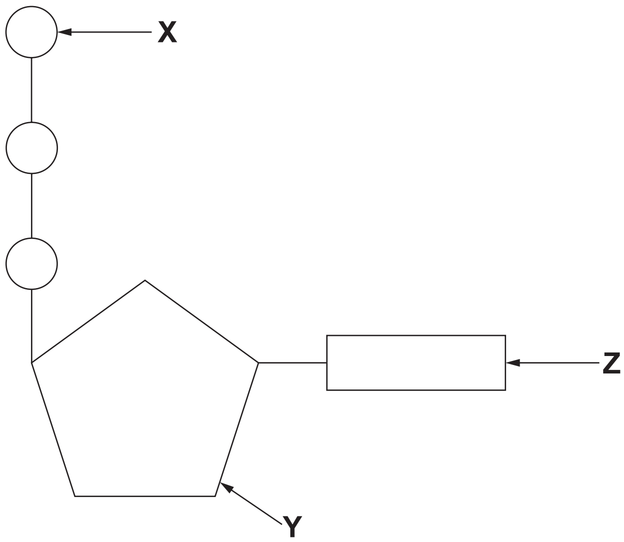
Examiner only

2. The diagram below shows a molecule of ATP which is produced during respiration and is an energy carrier molecule used in cells. When ATP is broken down into ADP and P_i, 30.6 kJ mol^{-1} of energy is released, which can be used for cellular activities.



(a) (i) State the names of the parts of the ATP molecule labelled **X**, **Y** and **Z**. [2]

X: ....

Y: ....

Z: ....

(ii) State **two** uses for the energy released from ATP in a plant cell. [1]

....

....

(b) One mole of glucose releases 2 870 kJ of energy when completely oxidised during aerobic respiration. Synthesis of one mole of ATP from ADP and P_i requires 30.6 kJ of energy. Calculate the percentage (%) efficiency of respiration if 38 moles of ATP are produced from one mole of glucose. Give your answer to three significant figures. [3]

% efficiency = ...

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